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Wake slow waves in focal human epilepsy impact network activity and cognition

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Abstract

Slow waves of neuronal activity are a fundamental component of sleep that are proposed to have homeostatic and restorative functions. Despite this, their interaction with pathology is unclear and there is only indirect evidence of their presence during wakefulness. Using intracortical recordings from the temporal lobe of 25 patients with epilepsy, we demonstrate the existence of local wake slow waves (LoWS) with key features of sleep slow waves, including the down-state of neuronal firing. LoWS were distinct from slow waves recorded in people without epilepsy and showed evidence of adaptation during periods of increased normal (cognitive task) and abnormal (interictal epileptiform discharges, IEDs) activity. Importantly, we also provide evidence that LoWS reduce increases in network excitability during IEDs but are associated with poorer memory. We therefore propose an “epilepsy homeostasis hypothesis”: that slow waves are activated in epilepsy to reduce aberrant activity at the price of transient cognitive impairment.

Introduction

During non-rapid eye movement (NREM) sleep, neurons undergo slow fluctuations of membrane potential, producing a burst firing mode (up-state) and a suppression of their activity (down-state). This fluctuation is reflected by slow oscillations (0.5-4 Hz) of the local field potential (LFP) termed slow wave activity (SWA, Fig. 1a). SWA is a canonical component of sleep [1,2] and has been proposed to be critical for sleep homeostasis [3–7] (specifically, the normalization of synaptic strength [6,8]), metabolic regulation [9], and metabolic waste clearance in general [10]. Elevated sleep pressure, as occurs after a wake period, is reflected by an increased rate, slope and amplitude of sleep slow waves (SWs), which decrease over the course of the night [4,11–16] (Fig. 1b). Furthermore, evidence suggests that neuronal synchrony, i.e., the concentration of neuronal firing during the up-state and neuronal silence during the down-state of sleep SWs, favours synaptic normalization [8,11,17], and is homeostatically modulated (high after a wake period, low after a sleep period, Fig. 1a) [12].

As occurs with wakefulness [4] and cognitive processing [3,18], epileptic activity results in increased synaptic connectivity [19–21] and metabolic need [22]. Consequently, the increased SWA observed during sleep after repetitive seizures has been interpreted as a compensatory mechanism [23–25] which counteracts the increased local metabolic demand, although there is currently no evidence for SW having a beneficial impact on epileptic activity. Furthermore, it is unclear whether such SWA might also appear during wakefulness, although this could constitute a mechanism to offset pathological activity, such as epileptiform discharges and seizures.

Here, using macro- and micro- electrode intracranial recordings from the temporal lobe of people with focal, pharmacoresistant epilepsy, we demonstrate the presence of highly local wake slow waves (LoWS) that recapitulate the core features of sleep slow waves, including the associated down-state of neuronal spiking activity. We hypothesized that these LoWS could serve a homeostatic purpose by normalizing

neuronal activity to prevent epileptic discharges, mirroring the function of sleep SWs which correct the excessive neuronal excitability accumulated during wakefulness that translates into high sleep pressure (i.e., sleep homeostasis [6]). If LoWS do serve a homeostatic function, then they should exhibit two key features: first, responding to increases in network activity associated with interictal epileptiform discharges (IEDs); and second, reducing abnormal activity linked with IEDs. In line with this hypothesis, we observed that progressive increases in neuronal activity (estimated by high-gamma (HG) power) before IEDs is paralleled by an adaptation of the slope and amplitude of LoWS, analogous to that observed in the response of sleep SW to increased sleep pressure, i.e., enhanced slope and amplitude [13,16]. Moreover, the longer the delay since the last LoWS, the higher the HG power during IEDs, suggesting that any protective function of LoWS dissipates with time. Lastly, we found that the higher rate of LoWS during an associative memory task was associated with longer reaction times (RT), supporting the prediction [5] that the substantial modulation of neuronal activity during SWs [12,26] impacts cognitive processing. Together, our findings indicate that LoWS with key features of sleep SW dynamically respond to changes in network excitability, reduce aberrant activity associated with IEDs, and impact on cognition. We therefore propose that LoWS represent a homeostatic process that comes at the cost of transient cognitive impairment.

Results

Identification of slow waves during wakefulness in patients with epilepsy

We first asked if we could reliably detect local slow waves during wakefulness in a cohort of people with focal epilepsy (n=17) undergoing intracranial EEG recordings, and then whether such slow waves had key features of sleep slow waves (Fig. 1). We used a validated algorithm for slow wave detection [13,26–28] to track their presence in the human hippocampus (mean ± SD of hippocampal contacts = 3.4 ± 1.8, see Extended Data Table 1), while people with focal epilepsy were performing an associative memory task. We excluded any slow waves associated with (i.e., within 1 s of) an IED and took several confirmatory steps to ensure that the identified slow waves were not mislabeled IEDs (see Online Methods).

The negative and positive components of slow waves, i.e., negative and positive slow waves, were identified (nSW and pSW respectively, Fig. 1c-d). To note, traces are inverted during acquisition, such that negative deflection (nSW) corresponds to positive extracellular potential, and vice versa. Rates of nSW and pSW were similar (effect size [ES] across electrodes expressed as mean difference, 95% CI: 0.002/s/electrode, -0.001 to 0.009, p=0.464, n=17 patients, Extended Data Fig. 1a) and were highly correlated across patients (Pearson correlation, r^2=0.79, p<0.0001, n=17, Extended Data Fig. 1b). nSW had a peak (extracellular) positive potential (Fig. 1c-d), which corresponds to a hyperpolarization of neurons. This inactive state is a core feature of, and defines, sleep SW [5,26,29]. For this reason, we subsequently focus only on the negative component of this slow wave activity (i.e., nSW), except where stated otherwise. nSW occurred at a mean ± SD rate of 0.06/s/electrode ± 0.02 (Extended Data Fig. 1a), well below the rate described during sleep [13,27,28]. The median oscillatory frequency of nSW, estimated by

the delay between the two 0-crossings, was 1.4 Hz (interquartile range, IQR: 1.1-1.7 Hz, n=17,712 hippocampal nSW identified across patients).

Negative slow waves during wakefulness share key characteristics with sleep slow waves

Next, we determined whether identified slow waves had key characteristics of sleep SW (Fig. 1). Analogous to sleep SW [26], nSW were associated with a down-state, as revealed by decreased high-gamma (HG, 45-130 Hz) power averaged between the two 0-crossings of the grand average nSW (-0.15 to 0.165 s around the trough of nSW versus baseline: ES, 95% CI: -0.137, -0.236 to -0.0884, p=0.0004, n=17, Fig. 1e). Using a time-resolved frequency analysis, we confirmed that the decrease in HG power was localized to the nSW trough (Fig. 1f). This was further confirmed through the observation of a decrease in HG power before the peak of pSW (from -0.42 to -0.13 s, Fig. 1g), the timing of which matched the peak of nSW occurrence before pSW in our data (around -0.36 s, Fig. 1g, i). Importantly, the peak of HG associated with IEDs was significantly higher than that during pSW (ES, 95% CI: 4,490 μV^2, 2,680-7,250, p=0.0008, Extended Data Fig. 2a), arguing against a mislabeling of the two events (see Online Methods for further confirmation analyses).

In the time domain, identified nSW occurred mainly as isolated events, with only 9-10% of nSW being preceded or followed by a pSW (with a delay of ± 0.37 s, Fig. 1h). The same analysis for pSW gave, as expected, a similar result (Fig. 1i). This suggests that nSW were mainly preceded and followed by positive deflections that either did not reach the amplitude or the frequency criteria for pSW (see Online Methods), and vice versa. In the spatial domain, nSW were mainly focal since the presence of a slow wave on one hippocampal contact was accompanied by a slow wave on only 27% of the other hippocampal contacts (Extended Data Fig. 3a). This microfocality was less pronounced in the amygdala (49%, Extended Data Fig. 3b) and more pronounced in the temporal neocortex (18%, Extended Data Fig. 3c), which could reflect differences in the volume and coverage of these brain regions. Similarly, the detection of a slow wave in the hippocampus, amygdala or temporal neocortex was accompanied, on average, by the detection of a slow wave in at least one of the other regions in only 24% of cases (Extended Data Fig. 3d), remarkably close to the percentage observed for sleep SW [26]. Slow wave incidence rates were similar in the hippocampus, amygdala, and temporal neocortex (Extended Data Fig. 3e). Given the highly focal nature of detected nSW, we subsequently refer to them as <u>lo</u>cal <u>w</u>ake <u>s</u>low waves (LoWS).

LoWS are associated with a down-state of neuronal firing, capturing a core feature of sleep slow waves

To confirm that the decrease in HG power during the LoWS trough corresponds to decreased neuronal activity, we examined LoWS in a separate cohort of patients who underwent hippocampal microelectrode implantation (n=8 patients) and identified unit spiking activity (see Online Methods and Fig. 2a). We observed a significant decrease in unit activity locked to the trough of LoWS (paired bootstrap statistic followed by false discovery rate correction across time bins, p<0.05 from -0.049 to 0.147 s, Fig. 2b). The maximal decrease of neuronal firing occurred just after the trough (Fig. 2a-d), as previously described for sleep slow waves in the hippocampus [26]. At this time point, > 50% of identified units in all 8 patients

showed a decrease in firing rate (mean ± SD = 78 ± 17%, one sample t-test, p=0.0024, Fig. 2c, n=8). Similarly, a comparison of normalized mean firing rates between the two 0-crossings of the grand average LoWS versus baseline confirmed that the decrease was significant (ES, 95% CI: -0.273, -0.455 to -0.0795, p=0.0368, n=8, Fig. 2e blue lines). Conversely, we found only non-significant trends in the modulation of neuronal activity around pSW (red lines in Fig. 2e) and IEDs (Extended Data Fig. 2b). In sum, neuronal activity shows a significant decrease shortly after the trough of LoWS during wakefulness, confirming the presence of this core feature of sleep slow waves in LoWS.

LoWS that occur in focal epilepsy are distinct from those in non-epileptic patients

We next determined if LoWS were unique to our cohort or also present in other cohorts of patients with focal epilepsy. We took advantage of an independent, publicly available dataset from patients with focal epilepsy undergoing intracranial recordings [30] (from Boran et al., 2020, referred to as the "Boran dataset" below) and detected similar wake slow waves in the hippocampus (see Online Methods), although they occurred less frequently (Boran dataset versus UCL dataset, ES, 95% CI: 0.0294/s/electrode, 0.0164-0.0398, p=0.0004, n=9 and 17 patients respectively, Extended Data Fig. 4a, c).

We further asked whether such LoWS were a feature of focal epilepsy or a ubiquitous phenomenon. Since it was not possible to obtain intracortical, mesial temporal lobe recording in healthy subjects, we examined a dataset of neocortical, electrocorticographic (ECoG) recordings in awake patients with Parkinson's disease [31] using the same detection algorithm. Slow waves in wakefulness were also detected in the Parkinson's disease patients but were rarer than in either epilepsy cohort (UCL dataset versus Parkinson's dataset, ES, 95% CI: 0.04/s/electrode, 0.03-0.05, p<0.0001, n=17 and n=16 patients respectively; Boran dataset versus Parkinson's dataset, ES, 95% CI: 0.01/s/electrode, 0.005-0.02, p=0.0138, n=9 and 16 patients respectively, Extended Data Fig. 4b-c).

Moreover, LoWS in the Parkinson's dataset were less easily detectable from background LFP (i.e., had a significantly lower signal to noise ratio) than those in either epilepsy cohort (Parkinson's dataset versus Boran dataset, ES, 95% CI: 1.05, 0.761-1.31, p<0.0001; Parkinson's dataset versus UCL dataset, ES, 95% CI: 1.15, 0.862-1.43, p<0.0001; Boran dataset versus UCL dataset, ES, 95% CI: 0.0988, -0.148 to 0.356, p=0.537, Extended Data Fig. 4d). Since slope and the associated down-state (i.e., mean HG power between the two 0-crossings of LoWS at -0.15 and 0.165 s) clearly distinguish LoWS from background noise, we used these intrinsic features to further characterize slow waves. When plotting all patients in a 2-dimensional space defined by these features, we observed that patients with Parkinson's disease constituted an isolated cluster with low slope and absent down-state. This was confirmed by statistical analysis of the inter-patient Euclidean distance within this 2-D space (see Extended Data Fig. 4e-f and Online Methods for further details and statistical analyses).

LoWS and interictal epileptiform discharges

There is burgeoning evidence of a complex interplay between epileptic and sleep-related network activity [23,27,32,33]. We thus examined the interaction between LoWS and interictal epileptiform discharges

(IEDs). We identified IEDs in all but one patient (mean rate ± SD across patients and electrodes: 0.01/s/electrode ± 0.01, n=16, Fig. 3a-c).

We next determined the rate of LoWS against that of IEDs. We observed a significant negative correlation across patients (Pearson correlation, r^2=0.51, p=0.002, n=16, Fig. 3d, see Supplementary Information for further discussion). We explored this further by determining whether this could explain why we found more LoWS in our dataset than in the other dataset of patients with epilepsy. After correcting the rate of LoWS by that of IEDs (see Online Methods), we indeed found a similar normalized rate of LoWS between both datasets of patients with epilepsy (Boran dataset versus UCL dataset, ES, 95% CI: -0.0004, -0.001 to 0.0001, p=0.22, Extended Data Fig. 4c, two last columns). Besides LoWS rate, neither the slope (Pearson correlation, r^2<0.001, p>0.99, n=16, Extended Data Fig. 5a), nor the amplitude (Pearson correlation, r^2<0.001, p=0.96, n=16, Extended Data Fig. 5b) of LoWS correlated with the IED rate.

Some evidence suggests that IEDs are preceded by a progressive increase in excitability [34,35]. In line with this, we observed a progressive increase in HG power prior to IEDs (β-estimate ± SD: increase of 1 log(μV^2) in HG power each 66.5 s ± 25.4, p=0.009, Fig. 3e). Thus, in parallel to this progressive increase in excitability before IEDs, we next asked whether LoWS undergo key electrophysiological changes similar to those of sleep SW in response to increased excitability after a period of wakefulness. Under high homeostatic sleep pressure, sleep SW present with high incidence rate, slope and amplitude [16] (see Fig. 1a-b). Using these features as covariates in a linear mixed model with time to the next IED as the dependent variable, we found that the slope (β-estimate ± SD: slope increases by 1% each 347 s ± 114, adj. p=0.0026, n=16, Fig. 4a, b) and amplitude (β-estimate ± SD: amplitude increases by 1% each 285 s ± 101, adj. p=0.01, n=16, Fig. 4a, c) but not the rate (β-estimate ± SD: rate changes by -1% each 55 s ± 96, adj. p=0.57, n=16) of LoWS increase significantly across time leading up to the next IED. Importantly, although these rates of change may appear small, they are more than double that reported in sleep SW across a whole night (at least for SW slope: change of 20-30% across 381 min of sleep) [13]. Thus, two classical markers of synaptic strength (the slope and amplitude of LoWS) [6] change in parallel to progressive increases in HG power prior to IEDs. This indicates that LoWS exhibit a key condition associated with homeostatic activity: they show adaptation in parallel to excitability, similar to the response of sleep SW to increased activity during wakefulness.

Effects of LoWS on IEDs

The results above indicate a response of LoWS to increases in network activity leading up to IEDs, but do LoWS also represent a mechanism to reduce these aberrant increases in network activity? IEDs are proposed to represent cellular depolarization [36] and can be generated locally or propagate from more distant sites. It has been suggested that the local or distal nature of IEDs can be determined by the presence or absence of an increase in gamma activity prior to the IED, respectively [34]. Moreover, high-gamma activity can further distinguish IEDs that are associated with increases in neuronal firing from those that are not [37,38].

To investigate the negative feed-back of LoWS on IEDs, we used a linear mixed model with delay since the last LoWS and the rate of LoWS in-between two successive IEDs as covariates and HG power during IEDs (from -0.05 to 0.05 s around peak amplitude) as the dependent variable. In addition, since we observed increases in HG prior to IEDs, we used the increase rate of baseline HG (ratio of HG power between -6 to -1 s before IEDs divided by -250 to -245 s) as a third covariate to identify any effects that might be specific to locally generated IEDs (i.e., those preceded by increases in HG), or that reflect an interaction between LoWS and background excitability. We hypothesized that LoWS would reduce increased excitability associated with IEDs (i.e., would impact the HG burst associated with IEDs) and would have greater effect on IEDs that are generated locally (i.e., the impact would be dependent on the level of baseline change in HG power). We found a significant effect of the delay since the last LoWS on HG power during IEDs (β-estimate ± SD: $7.7 \bullet 10^{-5} \pm 1.2 \bullet 10^{-5}$ log(μV^2/ μV^2)/s, $p=8 \bullet 10^{-11}$, blank red bar in Fig. 4d) and a significant interaction between baseline increases in HG power and delay since the last LoWS on HG power during IEDs (β-estimate ± SD: $1.8 \bullet 10^{-5} \pm 8.4 \bullet 10^{-6}$ /s, $p=0.031$, filled red bar in Fig. 4d). No significant association was found for LoWS rate prior to IEDs (Fig. 4d). Thus, a longer delay since the last LoWS is associated with high network excitability during IEDs (and vice versa), and this effect is more pronounced for IEDs that may be locally generated and / or associated with large preceding increases in background excitability. This suggests that LoWS also exhibit the second property expected of a homeostatic process: they reduce network excitability, similar to the proposed function of sleep SW.

Phasic modulation of LoWS during a cognitive task

Cognitive tasks are known to increase the brain's metabolic demand [39]. We therefore asked whether this increased demand also changed features of LoWS, consistent with a putative homeostatic role. To do so, we examined the properties of LoWS during the encoding and retrieval phases of a temporal lobe dependent associative memory task (see Online Methods). We found that the rate of LoWS increased during encoding (ES, 95% CI: 0.0116/s/electrode, 0.0038-0.026, $p=0.023$, n=17, Fig. 5a) but not retrieval (ES, 95% CI: 0.00446/s/electrode, -0.00343 to 0.0136, $p=0.348$, n=17, Fig. 5a). Furthermore, we observed that the slope of LoWS was steeper during both the encoding (ES, 95% CI: 0.06, 0.03-0.09, $p=0.002$, n=17, Fig. 5b) and retrieval (ES, 95% CI: 0.05, 0.02-0.08, $p=0.007$, n=17, Fig. 5b) phases of the memory task, while LoWS amplitude was higher during encoding (ES, 95% CI: 0.108 μV, 0.07-0.15, $p=0.0002$, n=17, Fig. 5c) but not retrieval (ES, 95% CI: -0.00104 μV, -0.0555 to 0.0459, $p=0.972$, n=17, Fig. 5c) phase. Together, these results support an adaptative modulation of LoWS by cognitive processes.

LoWS are associated with transient cognitive impairment

One reason why SW are believed to be restricted to sleep is their impact on neuronal activity, which might be expected to have a detrimental effect on cognition [5]. We tested this prediction by analyzing the relationship between LoWS and performance on the associative memory task. We found that the rate of LoWS during retrieval was significantly lower during trials with fast versus slow reaction times (RTs; ES, 95% CI: -0.11/s, -0.22 to -0.05, $p=0.0114$, n=17, Fig. 6a), suggesting that the occurrence of LoWS during retrieval impacts task performance. In contrast, the rate of LoWS was similar during good and bad

accuracy trials (Fig. 6b). Performing the same analysis on IEDs revealed no significant difference in the incidence rate between trials with fast versus slow RTs (Fig. 6c). However, the rate of IEDs was significantly lower during good versus bad accuracy trials in both the encoding and retrieval phases (ES, 95% CI during encoding: -0.04/s, -0.12 to -0.01, p=0.0114, n=15 subjects with ≥ 1 IED and ≥ 1 trial with bad and good accuracy; ES, 95% CI during retrieval: -0.02/s, -0.03 to -0.01, p=0.0032, n=16 subjects with ≥ 1 IED, Fig. 6d), consistent with previous studies [40]. Altogether, these results indicate that LoWS and IEDs are associated with distinct cognitive impairment: the former with increased RT and the latter with decreased accuracy.

Discussion

In sum, these findings reveal the existence of SW during wakefulness that recapitulate the distinctive features of sleep SW, including the down-state of neuronal activity. Although evidence supports a restorative function of SW during sleep [10,11], it was previously unknown whether such beneficial activity could also be activated during wakefulness in pathological conditions. Here, we provide evidence of changes in LoWS properties (slope and amplitude) before IEDs that reproduce the changes of sleep SW under high homeostatic sleep pressure [13,16]. We further show that LoWS occurring during this vulnerable period oppose the effect of baseline HG power in generating the burst of increased excitability associated with IEDs (i.e., elevated HG power during IEDs), suggesting that LoWS could operate as key components of epilepsy homeostasis [41], mirroring the well-known sleep homeostasis regulated by sleep SWs [6]. This is further supported by the negative correlation between LoWS rate and IED rate, which suggests that IEDs occur more frequently in patients with fewer protective LoWS. Last, we observed that these beneficial effects place a burden on cognitive processing, slowing reactions times in an associative memory task, consistent with their impact on neuronal activity [5].

While the finding of slow waves during wakefulness in an independent cohort of patients with Parkinson's disease suggests that they are not strictly specific to epilepsy, our work shows that they are more frequent and more readily distinguished from background in patients with epilepsy. Moreover, the wake slow waves in Parkinson's disease were distinct from LoWS in epilepsy as they had a reduced slope and were not clearly associated with a down state. It is worth noting, however, that the Parkinson's disease data [31] were obtained from neocortical surface recordings, as opposed to the mesial temporal, depth electrode recordings in both epilepsy cohorts. However, placing depth electrodes in the mesial temporal lobe of healthy subjects is not clinically justifiable.

Sleep SW have long been recognized and extensively investigated [4,6,10,12,16,18,28,29,42–44] whilst suggestive evidence of slow wave activity during wakefulness has only recently been described [45,46]. Previous studies of wake slow waves have made use of sleep-deprived animals [24], humans under sleep pressure undergoing scalp EEG recordings [45], or stimulation of cortical regions surrounding focal brain injury [25]. However, typical slow waves with an associated neuronal down-state have not previously been described in awake humans, and it was unknown whether this core feature of sleep SW occurred during

wakefulness. Indeed, the identification of neuronal down-states has been recognized as particularly challenging in awake humans, since these periods are expected to be very short [47]. However, by applying stringent criteria for selecting waves within the delta range, we have been able to characterize neuronal down-states during wakefulness in humans.

In the search for slow waves, some studies have included oscillatory activity up to 7-9 Hz [45,47–49], which increases the risk of incorporating theta oscillations that are typically enhanced by cognitive processes [50] or benign variants associated with drowsiness [51]. This is a concern, given our observation that the rate of LoWS increased with cognitive demand. This emphasises the importance of identifying neuronal down-states, in order to differentiate delta and theta oscillations from wake slow waves. The lower signal-to-noise ratio of scalp EEG also increases the difficulty of identifying wake slow waves, and the lower spatial resolution complicates the study of focal slow waves and their interaction with other local activity patterns, such as epileptiform discharges. Here, intracranial recordings allowed us to describe a local interplay between interictal epileptiform discharges and LoWS.

On a complementary note, many previous intracranial EEG studies describe separate “low” and “high” theta bands that are independently modulated by various cognitive processes [52–54]. However, the results presented here raise the possibility that changes in “low” theta power instead reflect an increased incidence of LoWS associated with cognitive effort. This would also explain why “low” theta is rarely observed in EEG or MEG recordings from healthy participants performing similar tasks [55] and incites further characterization of LoWS to clarify their interplay with cognitive processes.

The LoWS we observed in epilepsy resulted in decreased HG activity and neuronal firing akin to an interruption of network function, so it is not surprising that LoWS were also associated with slower memory retrieval. Importantly, this impact on cognition is distinct from the impaired accuracy associated with IEDs, which has been previously described [40]. The impact of LoWS on neuronal activity in the hippocampus could be less pronounced than that observed in other brain regions, as changes in the activity of hippocampal neurons during down- to up-state transitions are less evident than in neocortex and parahippocampus [56], despite similar phase locking to slow oscillations [26]. Furthermore, the normalization of neuronal excitability during sleep involves a complex dynamic affecting differentially neocortical cells with high- and low-firing rates [57], and further work is needed to clarify the effect of LoWS on neuronal subpopulations. It is also important to note that despite this negative short-term impact on cognition, the homeostatic role of LoWS is likely to be beneficial for longer-term brain function, similar to sleep slow waves [10,23]. In this context, the specificity of low-frequency (1-3 Hz) stimulation to elicit synaptic depression *in vitro* [58] and to decrease cortical excitability in humans [59] as well as its antiepileptogenic effect in a rodent model of epilepsy [60] fits well with the hypothetical antiepileptic property of LoWS. Our work proposes that the same pattern of activity is naturally employed by the brain in patients with epilepsy, and a failure to express LoWS could be detrimental. Modulating LoWS could thus represent an efficient neurostimulation protocol that enhances their protective activity to increase the epileptic threshold. Since preliminary works have shown that it is possible to experimentally amplify sleep

SW [61], the application or improvement [62] of such devices to the field of epilepsy could be within reach in the near future.

We observed changes in the properties of LoWS in the order of seconds to minutes before IEDs, suggesting that the time-scale for homeostatic control during wakefulness is shorter and more dynamic than that during sleep [23], as predicted by Vyazovskiy and Harris [5]. Consistent with this, our results argue in favor of discrete, isolated waves during wakefulness. This *phasic* homeostatic control, intimately linked to the occurrence of IEDs, could be explained by a temporal and spatial constrain on the generation of LoWS to minimize the deleterious effect of LoWS on cognitive functions [5]. Indeed, our data indicate that LoWS are a very local phenomenon that seems to affect mainly locally generated IEDs.

To summarize, our study provides evidence of “micro-sleep” modules activated during wakefulness in epilepsy that present with clear homeostatic features, showing not only adaptation in association with increases in network excitability before IEDs, but also negative feed-back on IEDs, and comes at the cost of cognitive lapses. Our work highlights the compromise made by a brain that is affected by transient pathological activity and calls for the development of interventional strategies aiming at promoting LoWS to test their direct therapeutic ability in epilepsy.

Online Methods

Patients

A total of 25 patients with medically refractory epilepsy (11 female, 23 right-handed, mean age of 38.5 yrs, 17 with macroelectrode recordings and 8 with microelectrode recordings, Extended Data Table 1) undergoing intracranial EEG monitoring for clinical purposes at the National Hospital for Neurology and Neurosurgery, London were included in this study. One supplementary patient with microelectrode recording was excluded because of high epileptic activity that would have impeded a reliable analysis of slow wave activity. Prior approval was granted by the NHS Research Ethics Committee (15/LO/1783), and informed written consent was obtained from each patient. Patients were recruited if the research protocol did not impact negatively on their clinical care, and if intracranial electrodes were targeted at mesial temporal lobe structures. Sex and gender were not considered in the study design, given the rarity of the data. Some of these patients participated in other research projects that have been published elsewhere [63,64]. Two publicly available datasets [30,31] relating to complementary cohorts from other centers (16 patients with Parkinson’s disease and 9 patients with epilepsy), were also included for control purposes. The Parkinson’s disease dataset [31] was published under a CC0 1.0 license (without copyright) and is available here: doi.org/10.7910/DVN/IO2FLM. The dataset of patients with epilepsy [30] was published under a CC BY-SA 4.0 license (https://creativecommons.org/licenses/by/4.0/) and is available here: https://doi.org/10.12751/g-node.d76994.

Task

A description of the associative memory task is provided in more detail elsewhere [63,64]. Briefly, the task was divided into encoding and retrieval phases. During encoding, participants were presented with 27 pairs of images that remained on screen for 6 s, preceded by a 2 s fixation cross and followed by a 2 s blank screen. Unbeknownst to participants, each pair was drawn from one of 9 discrete “events” comprised of a person, place and object. As such, each item (person, place, object) was paired with two other items over the course of encoding. During retrieval, participants were first presented with a single cue image for 3 s, drawn either from one of the 9 events viewed during encoding or an equal number of ‘New’ events, comprised of person, place and object images that had not been viewed during encoding. Next, participants were asked whether the cue image was “Old or New?” – i.e., whether that image had formed part of a pair viewed during encoding or not. If the cue image was ‘Old’, whether or not the participant correctly identified it as such, they were then successively asked to indicate which two images it had been paired with, from two groups of four alternative choices drawn from the same image category across all ‘Old’ events (i.e. if cued with a person, they were first asked which place and then which object it had been paired with during encoding). In each case, participants had unlimited time to indicate their response with a key press, and reaction times (RTs) were recorded.

To test if the rate of IEDs or LoWS during encoding was associated with changes in accuracy or RTs, the mean incidence rate across the two 6 s encoding trials where each image was presented was measured. To test if the rate of IEDs or LoWS during retrieval was associated with changes in accuracy or RTs, the incidence rate during the time taken to make an associative memory response was measured.

Data acquisition

Macroelectrode EEG data were recorded continuously at a sample rate of 2048 Hz (Patients 12 and 13), 512 Hz (Patients 1 and 16), or 1024 Hz (all other patients) using a Micromed SD long-term monitoring system (Micromed). Recordings made at a higher sampling rate were downsampled to 512 Hz before any analyses were performed. Microwire EEG data were recorded continuously at a sample rate of 30 kHz using a Blackrock Neuroport system (Blackrock Neurotech LLC).

Data analysis

Code and software used for analyses

Analyses were performed using custom Matlab (The MathWorks, v. 2022a) code and the Fieldtrip toolbox [65]. Statistical analyses were performed using the web application EstimationStats [66], Graphpad Prism (v. 9.5.0), SPSS (v. 29.0.0.0) and Matlab (v. 2022a). Figures were designed in Matlab, Graphpad Prism and Adobe Illustrator.

Identification of IEDs

IEDs were identified in a semi-automated manner. First, the data were screened using a published algorithm for rodent models of epilepsy [32] that was adapted to human recordings. Briefly, data were

filtered between 20-80 Hz (using a second order Butterworth filter) and the amplitude of the filtered signal estimated using the absolute value of the Hilbert transform. Candidate IEDs were then defined as periods where both the amplitude of the filtered signal exceeded 3.5 times the mean envelope and the amplitude of the unfiltered signal exceeded 4 times the mean envelope. A refractory period of 1 s after each candidate IED was added, during which no other IED could be detected. Next, if >1 recording contact in any single region of interest (hippocampus, amygdala, temporal neocortex) exhibited a candidate IED within a window of ±0.3 s, only the candidate IED with the highest amplitude was retained. Finally, all candidate IEDs were curated by visual inspection and missed IEDs were manually added.

Identification of slow waves

Slow waves were detected using a standard algorithm [13,26–28]. First, the LFP signal was filtered between 0.5-4 Hz (using a second order Butterworth filter), all positive-to-negative zero-crossings and negative-to-positive zero-crossings in the filtered signal were detected, and events with a duration of 0.25 to 1 s were selected as candidate slow waves. Next, any candidate slow waves that occurred ≤1 s after a detected IED on any of the recording contacts in the same brain region were discarded. Finally, only the remaining candidate slow waves whose amplitude exceeded the 90th percentile were retained. Those slow waves were then visually inspected, and any that appeared artifactual or had an apparent IED occurring shortly beforehand were discarded. We were concerned by the fact that IEDs are typically followed by a slow wave, but it was not possible to visually inspect all slow waves, due to the large number detected. In addition to the precautionary measures taken above, we thus undertook several verification steps to ensure that detected slow waves were not mislabeled IEDs:

1. Detected slow waves were visually similar to those described in other publications (Fig. 1c-d) [13,28].
2. Detected slow waves and IEDs had a different spectral signature (Fig. 1f-g and Fig. 3c).
3. 45-130 Hz high-gamma (HG) power during IEDs (measured either during the time window used for down-state analysis, -0.15 to 0.165 s; or during the time window of IED peak, -0.05 to 0.05 s; was significantly higher than during pSW (ES, 95% CI: $4.49 \bullet 10^3$ μV^2, $2.68 \bullet 10^3$-$7.25 \bullet 10^3$; and ES, 95% CI: $2.24 \bullet 10^4$ μV^2, $1.29 \bullet 10^4$-$3.71 \bullet 10^4$, p=0.0008 for both, Extended Data Fig. 2a). Note that since HG power is decreased during LoWS, we did not compare it with HG increases during IEDs.
4. If slow waves were mislabeled IEDs, then variability in the rate of slow waves across electrodes should follow variability in the rate of IEDs rate across the same electrodes. To test this, we defined an index of 'parallel rate variability' (PRV) that records the ratio of IED rate to slow wave rate for each electrode minus the ratio of the sum of IED rate and slow wave rate across all electrodes for each participant. As such, the median PRV value is expected to be significantly different from 0 if the rates of IEDs and slow waves do not follow the same pattern across electrodes. The PRV is defined by:

$$prv = \sum_{n=1}^{N} \left(\frac{rate(ied)_{e_n}}{rate(sw)_{e_n}} - \frac{rate(ied)_{e_n} + rate(ied)_{e_{n+1}} + rate(ied)_{e_N}}{rate(sw)_{e_n} + rate(sw)_{e_{n+1}} + rate(sw)_{e_N}} \right)^2$$

Where n=1 to N indexes the electrodes in a specific patient. Across participants, the median PRV was

significantly different from 0 (median, 25-75th percentile: 0.06, 0.001-1.62, one sample Wilcoxon test, p<0.0001, Extended Data Fig. 6), indicating that variability in the rate of IEDs rate is not paralleled by variability in the rate of slow waves across electrodes. We calculated the PRV only in the hippocampus, where all main analyses were performed.

5. If LoWS were mislabeled IEDs, then electrodes that exhibit the greatest number of IEDs should also record the greatest number of slow waves. However, there was no difference in the rate of slow waves rate between all electrodes but the one with the highest rate of IEDs rate and that with the highest rate of IEDs (dES, 95% CI: -0.002/s, -0.01 to 0.003, p=0.595, Extended Data Fig. 7).
6. LoWS and IEDs have differential impacts on memory processing (Fig. 6).
7. LoWS and IEDs rates are negatively correlated (Fig. 3d). If LoWS were a subpopulation of mislabeled IEDs, and since LoWS detection is semi-automatic (except for the visual removal of artefacts and post-IED waves), then the proportion of mislabeled IEDs as LoWS should be constant across participants, resulting in a positive correlation (we provide further discussion of this point, as well as statistical confirmation, in the Supplementary Information).
8. Finally, we also wished to address the possibility that the slow waves we identified corresponded to low frequency post-IED waves that propagate more readily across electrode contacts than the sharp component of the IED. To test this, we measured the absolute amplitude of the raw LFP signal at the peak of each IED (i.e., the amplitude of the sharp component) across all hippocampal electrodes and the maximum absolute amplitude in the 0 to 0.5 s window after that peak (i.e., the amplitude of the subsequent slow wave) and then quantified the relative reduction in amplitude of each with distance from the focus. Across 7 adjacent recording contacts (the maximum number available on each depth electrode), there was no significant difference in these normalized amplitudes (Extended Data Fig. 8a-b). Since our main focus was on the deepest (i.e., most medial) recording contacts (located in the hippocampus), we also compared the ratio of amplitudes on the 1st electrode (electrode of interest) and 2nd (adjacent electrode contact) and did not find a significant difference (dES, 95% CI: 0.0391, -0.0513 to 0.153, p=0.49, Extended Data Fig. 8c). In sum, this indicates that the amplitude of post-IED slow waves decreases with distance from the focus at the same rate as the sharp component, arguing against the wider propagation of post-IED slow waves compared to the sharp component which could make post-IED waves appear as LoWS (i.e., without the sharp component).

Corrected rate of slow waves. Since we observed a negative correlation between the rate of LoWS and IEDs (Fig. 3d), we computed the rate of LoWS corrected for the rate of IEDs to verify whether variability in the rate of IEDs could explain discrepancies between our dataset and the Boran dataset (Extended Data Fig. 4c, 3 first columns). The corrected rate was calculated by multiplying the LoWS rate by the IED rate (Extended Data Fig. 4c, 2 last columns), to account for the negative correlation between these variables.

Signal-to-noise ratio of slow waves. Before calculating the signal-to-noise ratio and to approximately match our dataset, data from the Parkinson's and Boran datasets were downsampled to 500 Hz from their original sampling frequency of 1000 Hz and 2000 Hz, respectively. For each dataset, we then normalized the amplitude of the signal in a ± 3 s around each slow wave to the maximal amplitude in the

same time window. The signal-to-noise ratio was then measured as the ratio of the peak amplitude of the wave to the standard deviation across the entire window.

Identification of slow waves in the Boran dataset. We used another dataset of patients with epilepsy recorded with intracortical electrodes [30] to assess the reproducibility of slow waves detection. To identify hippocampal slow waves, we used the Juelich Histological Atlas, which computes a probability that each contact is located in a subset of brain regions. We considered a contact to locate in the hippocampus if the hippocampus was the most probable region and the probability was ≥ 25%. We used the Matlab *mni2atlas.m* functionfor that purpose (https://github.com/dmascali/mni2atlas/releases/tag/1.1).

Identification of slow waves in patients with Parkinson's disease. We followed the same procedure as the one described for the detection of slow waves in our dataset, except that we did not identify IEDs. None of the 16 patients were suffering from epilepsy (personal communication with Dr. Merk). Data were converted into µV for display. We only used the signal from neocortical contacts.

Interactions between IEDs and LoWS

LoWS-rate around IEDs. To measure variability in the incidence rate of LoWS around IEDs, we calculated the mean rate of LoWS across electrodes in a series of non-overlapping 5 s time windows relative to the closest IED (up to 250 s before IEDs). These values were then normalized by the maximal LoWS rate for each patient. We then used the software SPSS to compute linear mixed models using time as the dependent variable and rate, slope and amplitude of LoWS as covariates (see below). The y-intercept was specific to each subject.

LoWS-slope around IEDs. To measure variability in the slope of LoWS around IEDs, we computed the average slope of all LoWS in each 5 s time window relative to the closest IED (up to 250 s before IEDs). To do so, we identified the minimum of the 0.5-4 Hz filtered signal in a 1 s window following LoWS onset and defined LoWS slope as the ratio between the amplitude and post-onset time of this minima. Each slope was normalized by the value of the highest absolute slope for that patient, so that all values lay in the range of 0 to 1. These values were then used as covariate in the linear mixed model, as indicated above.

Amplitude of LoWS around IEDs. To measure variability in the amplitude of LoWS around IEDs, we followed the same procedure described above but using the absolute minima of the 0.5-4 Hz filtered signal in a 1 s window following LoWS onset, in place of the estimate of LoWS slope.

HG power before IEDs. We used the Fieldtrip [65] function *ft_freqanalysis.m* to calculate the HG power (45-130 Hz) during non-overlapping windows of 1 s before IEDs (multitaper frequency transformation) and log-transformed the result. Each IED presented with a different pre-IED time-window, which length depended on the delay since the last IED. Since frequency power is exponentially distributed, it was important to include as many trials in each non-overlapping time-bin before IEDs, to avoid that very low, or very high, values of HG power affect differentially time-bins with only a few or many trials. Hence, we

selected the same number of trials for each time-bin prior to IEDs by identifying for each participant the maximal pre-IED duration that would encompass at least 50% of all IEDs. HG power was then used as a covariate and time to the next IED as the dependent variable in a mixed linear model.

Interaction between LoWS, baseline HG and HG power during IEDs

To calculate the increase rate of HG power before IEDs, we measured the ratio of HG power (45-130 Hz) before IEDs (-6 to -1 s before IEDs) to a period away from IEDs (-250 to -245 s) using *ft_freqanalysis.m* (multitaper frequency transformation) and log-transformed the result. The period (-250 to -245 s) and the size of the time-windows (5 s) were chosen for homogeneity with the analysis of rate, slope and amplitude before IEDs. To calculate HG power during IEDs, we measured the HG power during IEDs (-0.05 to 0.05 s around IEDs, thus not overlapping the window for increase rate of baseline HG) and normalized it against HG power sampled randomly during artefact-free periods. The delay since the last LoWS was recorded in seconds. The rate of LoWS was calculated as the number of LoWS between the last IED and the IED of interest, divided by the duration of that interval (in seconds). Then, we used a linear mixed model with HG power during the IED as the dependent variable and delay since the last LoWS, rate of LoWS, interaction [baseline increases in HG*delay] and [baseline increases HG*rate] as covariates. We did not include “baseline increases in HG” as an independent covariate, since it can be expected that an increase in HG power will be followed by high values of HG, and vice versa. We selected HG power during IEDs, rather than the peak amplitude of IED, because the peak amplitude is sampled at only one timeframe, which would not reliably represent the relatively long duration of IEDs (see Fig. 3b), and because the peak amplitude can vary with shifts in the ongoing baseline activity. Furthermore, HG activity is typically considered a reliable proxy of neuronal activity [38].

Time-frequency analyses

To measure changes in 45-130 Hz HG power associated with LoWS (Fig. 1e), we used Fieldtrip [65] function *ft_freqanalysis.m* (multitaper frequency transformation) for the time window between the two zero-crossings of the grand average LoWS (-0.15 to 0.165 s around the LoWS trough). Power values were baseline corrected using average power at each frequency across the -2.9 to -0.2 s and 0.2 to 2.9 s time windows relative to the LoWS trough. As a control, the same analysis was performed relative to a randomly sampled set of time bins (as opposed to the occurrence of genuine LoWS). Importantly, changes in HG power associated with pSW and IEDs were not baseline corrected, because changes in HG power were observed within the baseline time windows (e.g., during the down-state before pSW [Fig. 1g] and following IEDs [Fig. 3c]). For the time-frequency decompositions in Fig. 1f,g (LoWS) and 3c (IEDs), dynamic changes in oscillatory power were estimated using a seven-cycle Morlet wavelet transform (implemented using the Fieldtrip function *ft_freqanalysis.m*) in ± 3 s time windows around IEDs and LoWS across a 1.5 to 130 Hz frequency range. For control purposes, we used the functions *randsample.m* or *datasample.m* to randomly generate twice as many time bins (for LoWS) or the same number of time bins (for IEDs), excluding artefactual periods and time windows of ± 3 s around another IED or LoWS (for LoWS) or time windows of ± 3 s around another IED (for IEDs).

Micro-electrode recordings

Single and multi-unit activity was identified using the wave_clus toolbox [67]. Briefly, data were filtered in the 300-3000 Hz range and events whose amplitude was within 4-50 standard deviations of the median noise identified as candidate spikes. Spikes were then clustered automatically based on wavelet features of the waveform before single- and multi-unit activity clusters (denoted together as unit spiking activity or "units" in the manuscript) were manually identified based on the inter-spike interval distribution and average waveform. The firing rate of each unit was then convolved with a 0.2 s Gaussian window and normalized to the mean firing rate across epochs (± 3 s around individual LoWS). Finally, units with a mean firing rate < 2 Hz across the entire recording session were discarded, following [26]. The statistical analysis in Fig. 2c was performed with units that had a mean firing rate > 2 Hz across the entire EEG and with units that had a mean firing rate > 2 Hz during the surrounding baseline (-2.9 to -0.2 and 0.2 to 2.9) and both came out as significant after Bonferroni-Holm correction. For display (Fig. 2d) we used units with mean firing > 2 Hz during the surrounding baseline.

To obtain a time-resolved evaluation of the firing rate around LoWS, we first obtained the mean firing rate per patient (n=8) and downsampled these time series to 10 Hz, thus obtaining a resolution of 0.1 s for statistical analysis. We then used a bootstrapping approach as follows: for each time bin from -1 to 1 s around the LoWS trough, the difference between the mean firing rate around LoWS and five times as many randomly sampled time bins was calculated. Across 5000 bootstraps, the identity of these conditions was then randomly switched to obtain a null distribution of paired firing rate differences. We then identified where the actual firing rate difference lay on the bootstrapped distribution and recorded the percentile to calculate a two-tailed *p*-value. We chose more control markers to increase the reliability of baseline estimation. Finally, *p*-values were adjusted by the false-discovery rate (function *fdr_bh.m* [68]).

Mean firing rates during the down-state were computed between the two zero-crossings of the mean LoWS (from -0.15 to 0.165 s around the LoWS trough) and normalized by the mean firing rate during the surrounding baseline window (-2.9 to -0.2 and 0.2 to 2.9 s around the LoWS trough). For statistical analysis, these values were compared against the firing rate occurring around nSW and pSW, which occurrence had been shuffled (left end of each line in Fig. 2e).

Analysis of IEDs, LoWS and memory performance

To assess the effect of IEDs and LoWS on memory performance, we tested whether mean incidence rates during encoding (i.e., during the 6 s of image presentation) and retrieval (i.e., during the variable duration period between the associative memory question appearing and participant's response) differed between good and bad accuracy trials (i.e. those with correct versus incorrect responses) and fast vs slow RT trials (i.e. those with RTs below the 20th percentile and above the 80th percentile across all trials).

Statistical analyses

Using the web application estimationstats.com [66], we performed estimation statistics rather than significance testing for most statistical analyses. Estimation statistics provide numerous advantages [69], one of them being that the statistical analyses relies on an evaluation of the effect size. For these analyses, the *p*-values displayed represent the probability to obtain the same effect size if the mean paired difference is equal to 0. We indicate the mean paired difference (effect size, ES) as well as the 95% confidence interval (95% CI). *P*-values are obtained through two-sided permutation t-tests using 5000 bootstraps. Since estimation statistics are not null-hypothesis testing, they do not require correction for multiple comparisons. Even so, we checked whether *p*-values resisted a Bonferroni-Holm correction when multiple comparisons were applied: % of unit spiking activity showing a decrease during down-state; LoWS rate across 3 datasets; LoWS signal-to-noise ratio; analysis of cluster (Extended Data Fig. 4f); LoWS effect on behavior; IEDs effect on behavior; LoWS slope during encoding and retrieval; LoWS amplitude during encoding and retrieval; LoWS rate during encoding and retrieval. We found that all *p*-values remained significant. When multiple comparisons were applied on null-hypothesis statistics, we also applied a Bonferroni-Holm correction. For most plots, we display the effect size on the right of the respective panel.

To test whether the waves identified in patients with Parkinson's disease [31], in another dataset of patients with epilepsy [30], and in our dataset, were similar or constituted separate clusters, we plotted all participants' average slow wave in a 2-dimensional space defined by slow waves slope and down-state (estimated by HG power). We then measured the Euclidean distance between two points (slow waves in the 2D space defined above) of the group of interest, which we call distance "a". We then calculated the Euclidean distance between one point of the group of interest and one point sampled across all points (for patients with Parkinson's disease) and one point of the group of interest and one point sampled across all patients with epilepsy (for patients with epilepsy), which we call distance "b". Each of these points was randomly sampled 1000 times. We then calculated the difference between distance "a" and distance "b" across these 1000 samples. Across these 1000 differences, we then calculated the z-value of a one-sample Wilcoxon signed rank test (*signrank.m*) and the median of the difference, the null hypothesis being a median equal to 0 (the distance between two points within one group is equal to the distance across groups). We repeated this operation 1000 times to obtain 1000 z-values and 1000 medians. We then calculated the corresponding two-tailed *p*-value and the mean median of the distance difference. We used two sets of 1000 repetitions as the one-sample Wilcoxon signed rank test tends to virtually always gives a significant *p*-value for large datasets, even for very small differences that cast doubt on the meaning of the observed effect (risk of type I error). On the other hand, datasets that are too small fail to reliably assess a potential effect (risk of type II error). Hence, by extracting one z-value based on 1000 repetitions, we mitigate the risk of type I error and by calculating the mean z-value across 1000 repetitions, we mitigate the risk of type II error. To test whether the subset of patients with Parkinson's disease formed an isolated dataset, we followed this strategy:

1. We tested whether patients with epilepsy from the Boran dataset [30] presented with waves that were closer to themselves than to a group composed of waves from the Boran dataset and ours. This

analysis revealed a significant positive difference of distance (median of the difference, 25-75th percentile: 0.04, 0.03-0.06, adj. p<0.0001, Extended Data Fig. 4f), indicating that patients from the Boran dataset are more distributed than all patients with epilepsy taken together. The inverse, i.e., whether our cohort was isolated from the Boran dataset, showed no significantly different distance (median, 25-75th percentile: -0.008, -0.01 to -0.0002, adj. p=0.1165, Extended Data Fig. 4f). Altogether, this shows that patients with epilepsy from the Boran dataset and ours are intermingled.

2. We then tested whether patients with Parkinson's disease [31] presented with waves that are closer to themselves than to a group composed of all patients together. This revealed a significantly negative difference (median, 25-75th percentile: -0.04, -0.05 to -0.04, adj. p=0.0004), indicating that slow waves from patients with Parkinson's disease indeed constitute a separate population.

Thus, slow waves identified in patients with Parkinson's disease constitute a separate population from those identified in two independent datasets of patients with epilepsy.

Abbreviations

HG: high-gamma

IEDs: interictal epileptiform discharges

LoWS: local wake slow wave

NREM: non-rapid eye movement (sleep)

nSW: negative slow wave, i.e., associated with down-state of neuronal activity

pSW: positive slow wave, i.e., associated with up-state of neuronal activity

SW: sleep slow wave

SWA: slow-wave activity (power between 0.5-4 Hz)

Declarations

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None.

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Custom-codes and data are available upon reasonable request to the corresponding author.

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Figures

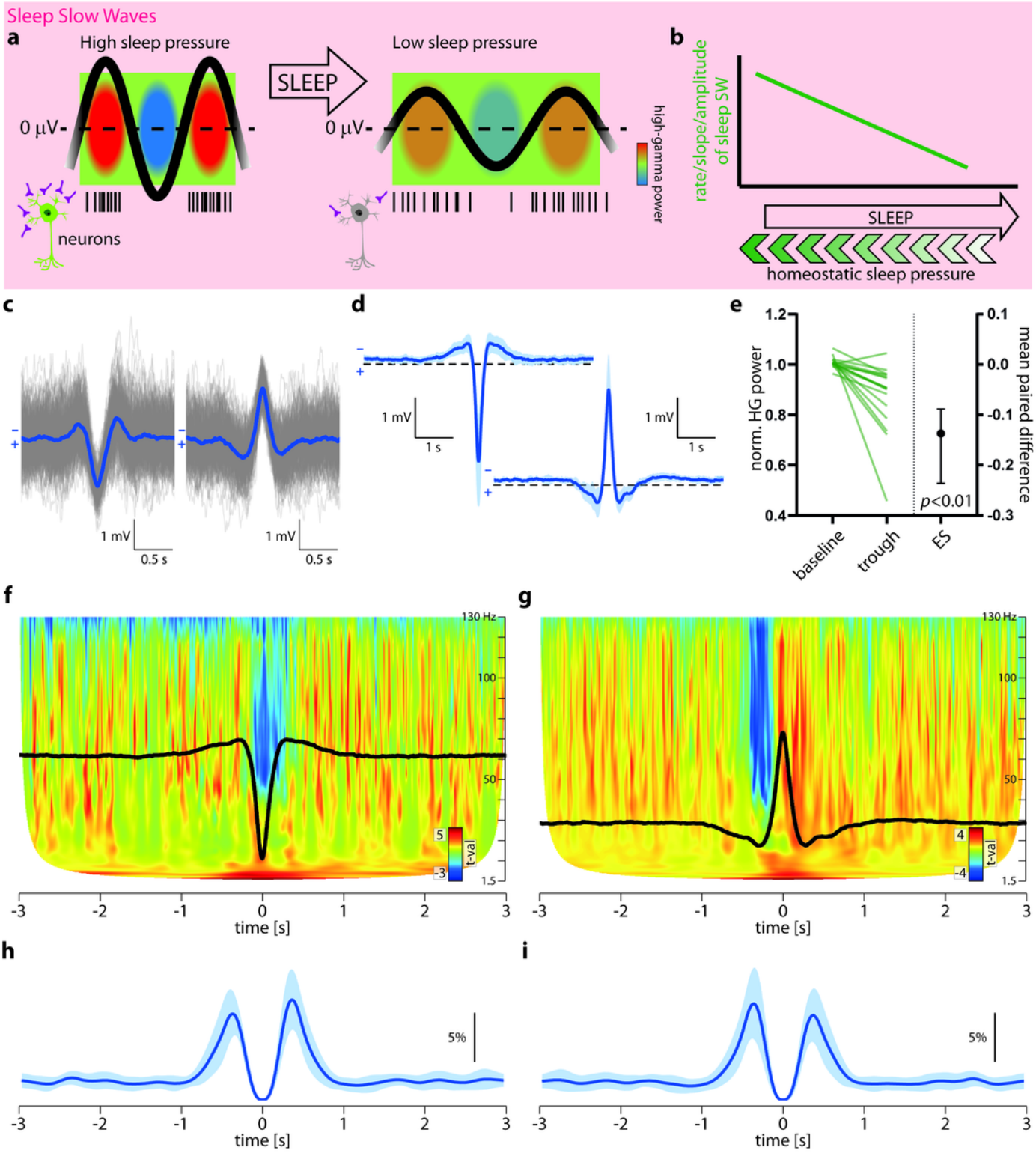

Figure 1

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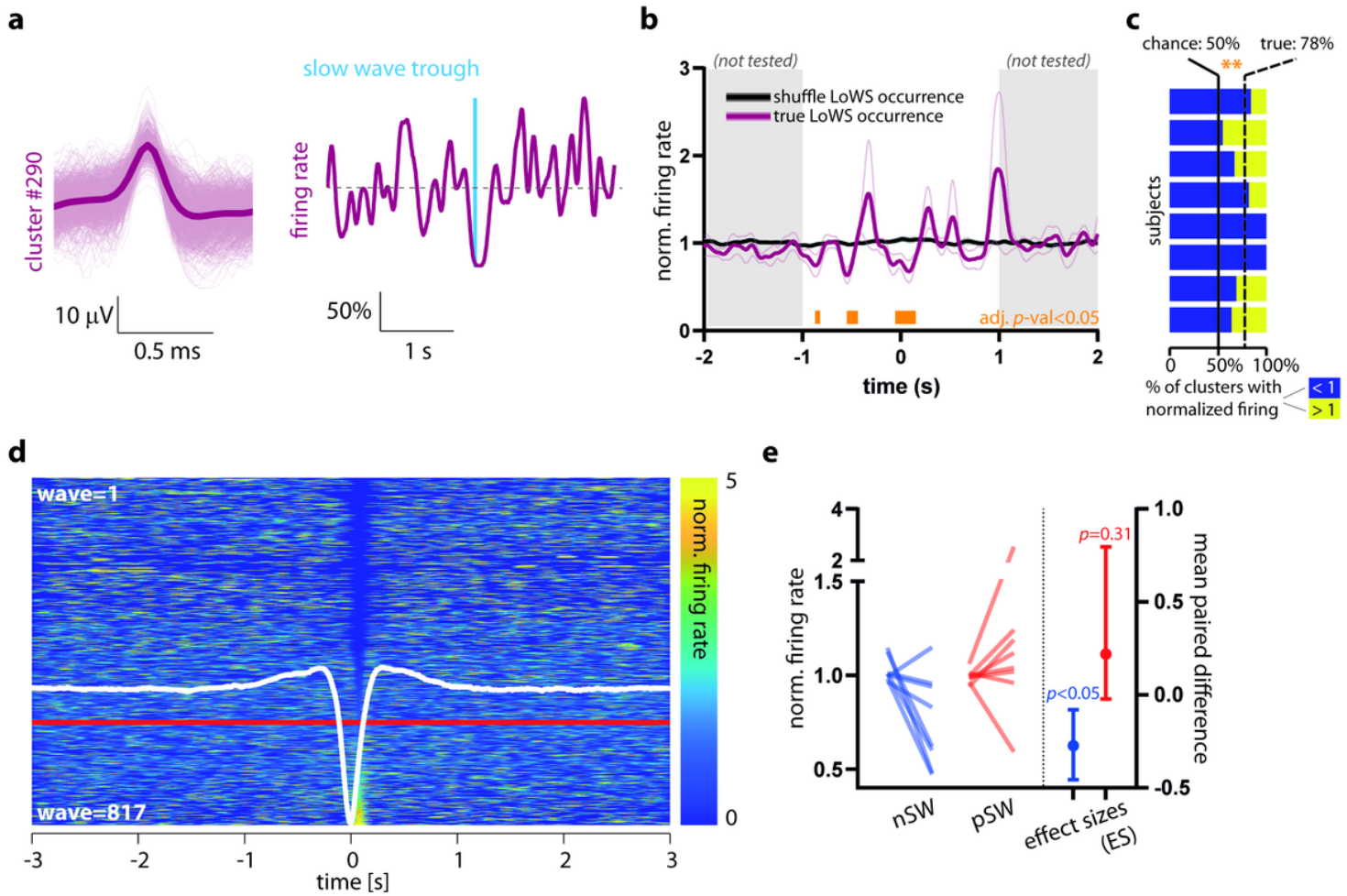

Figure 2

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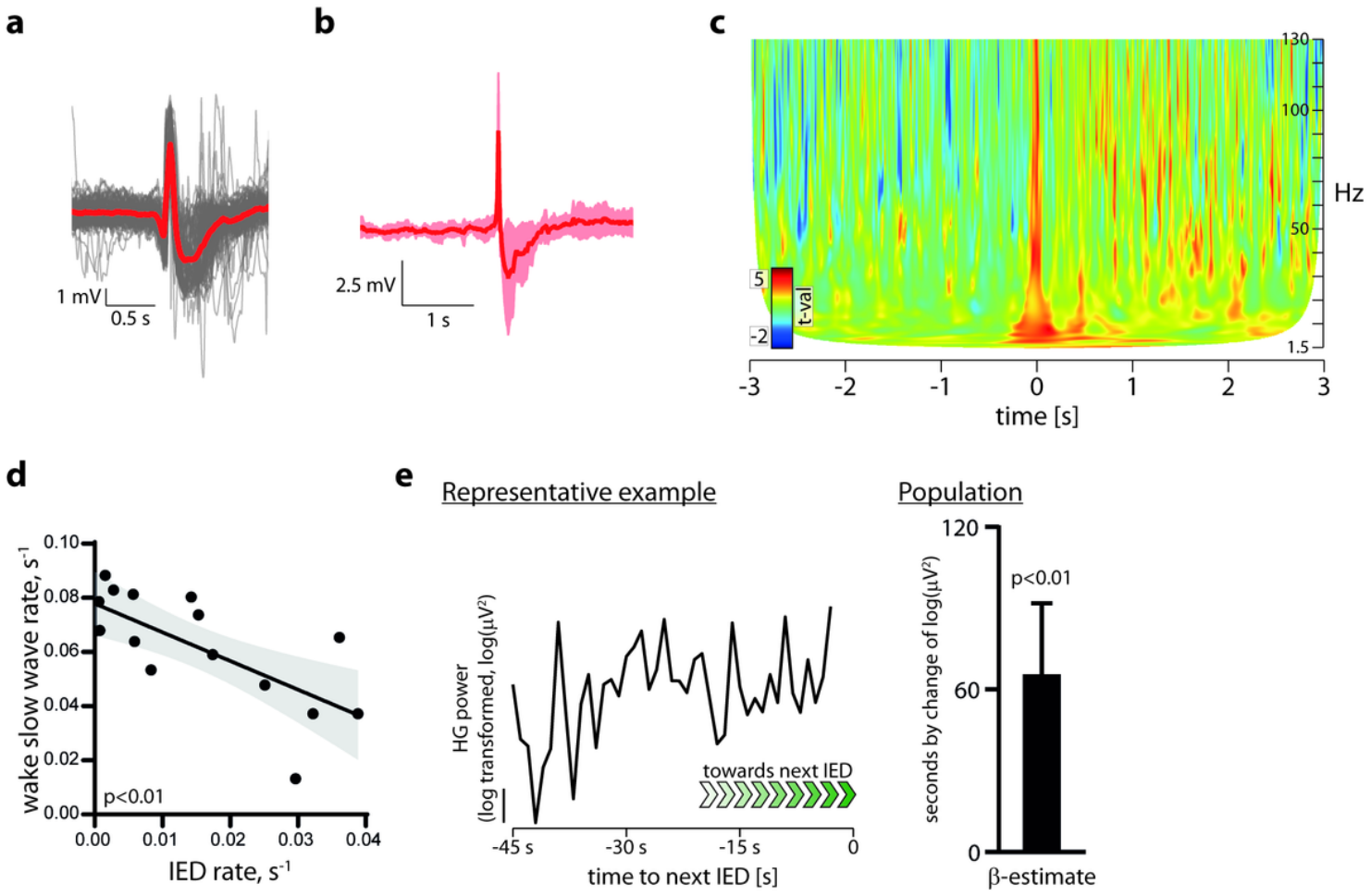

Figure 3

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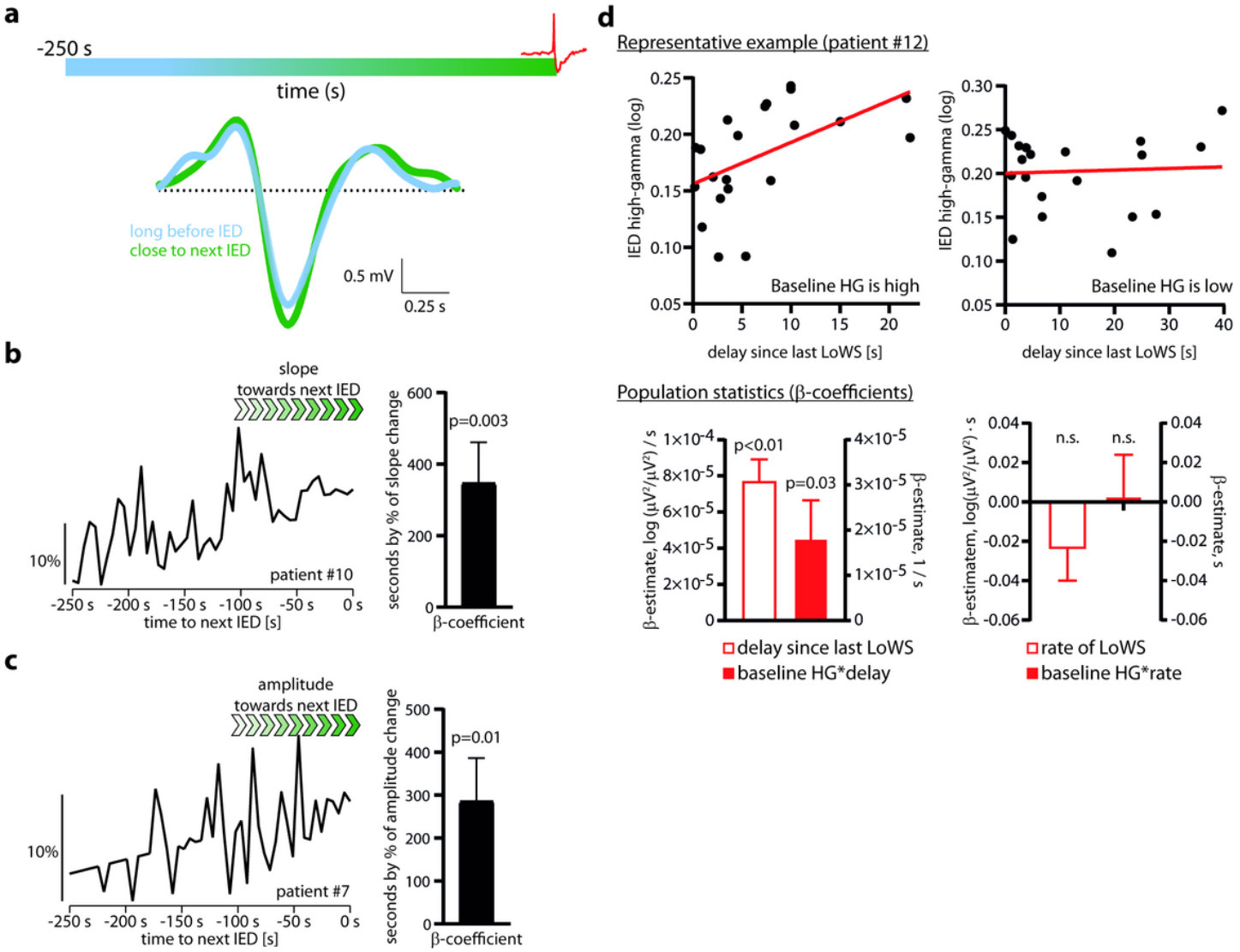

Figure 4

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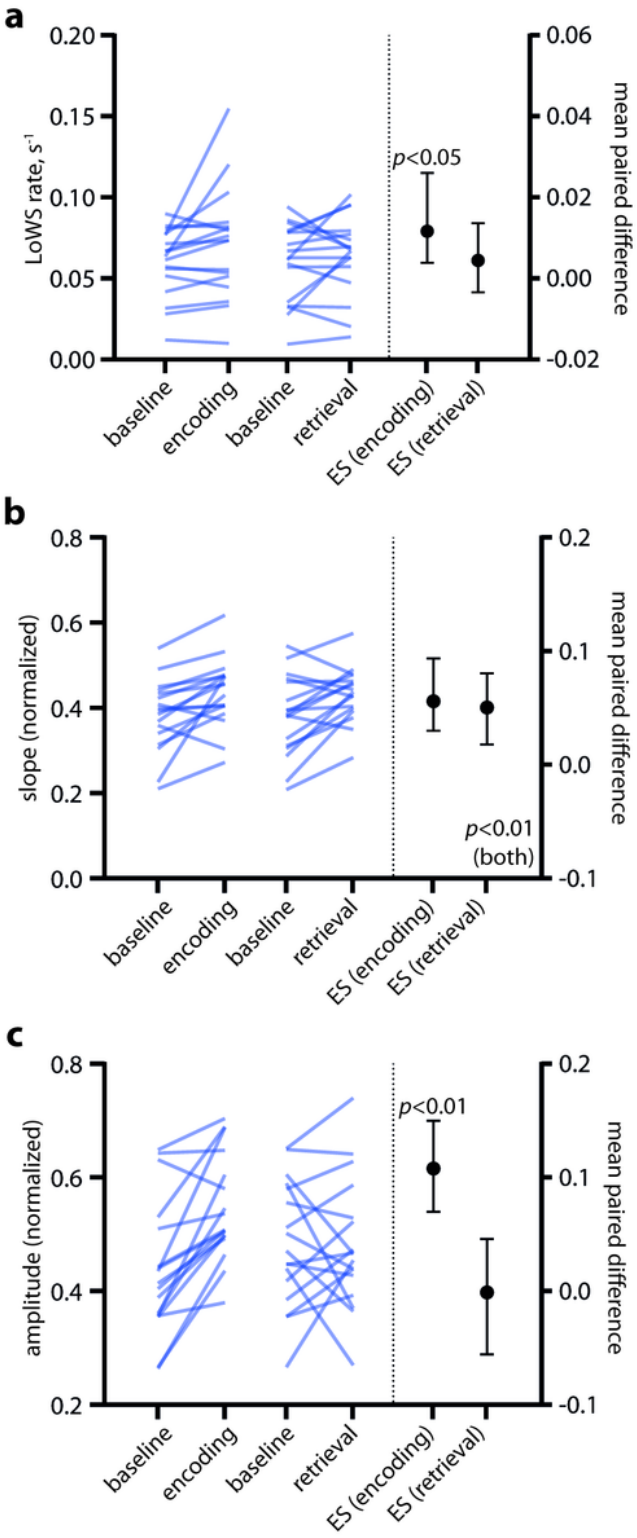

Figure 5

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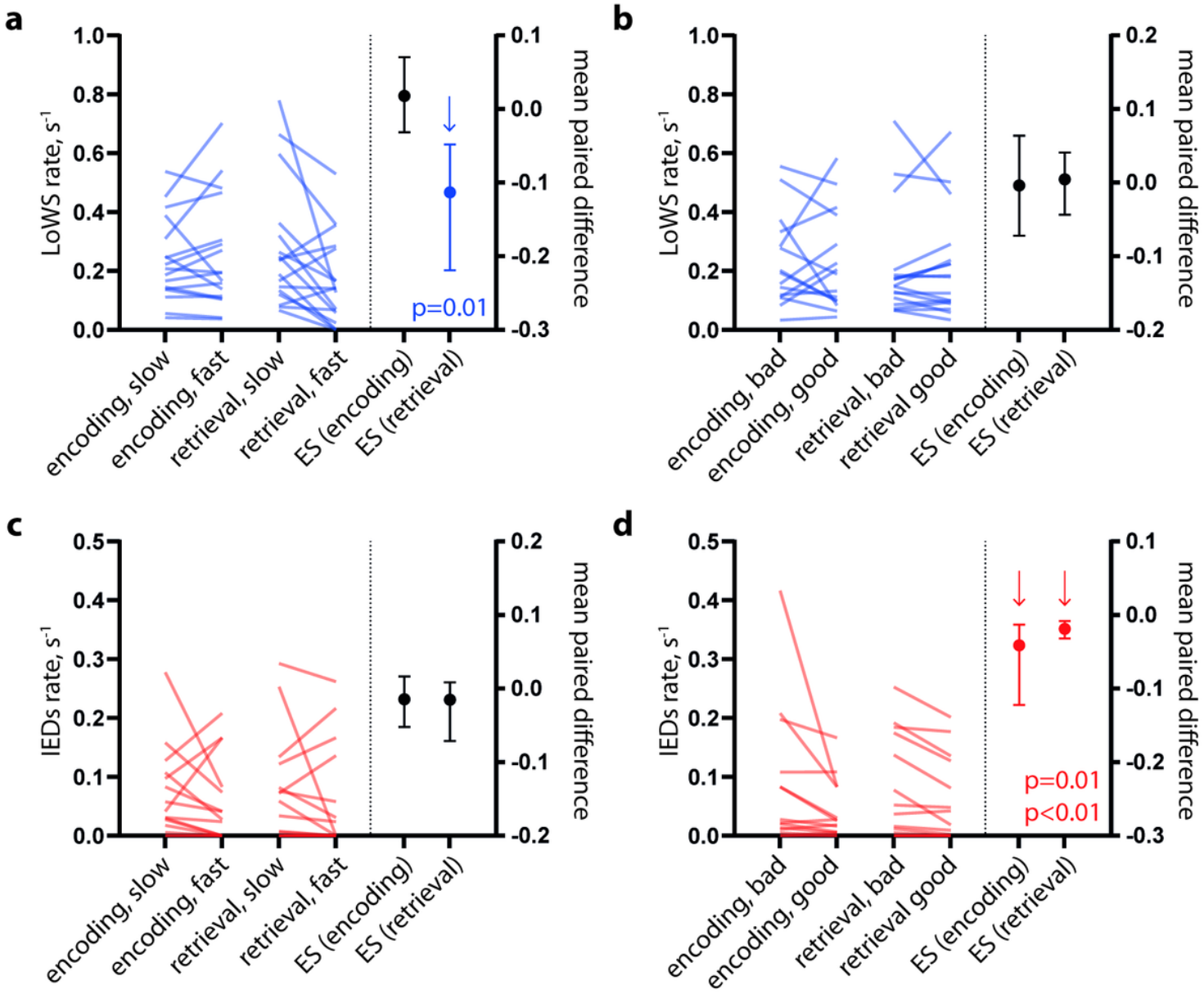

Figure 6

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